Geometry Problem Set 1

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Problem 1. (Centroamerican and Caribbean 2020 P4) Consider a triangle ABC with $BC > AC$. The circle with center C and radius AC intersects the segment BC in D. Let I be the incenter of triangle ABC and Γ be the circle that passes through I and is tangent to the line CA at A. The line AB and Γ intersect at a point F with $F \neq A$. Prove that $BF = BD$.

Problem 2. (APMO 2020 P1) Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC. The tangent to Γ at A intersects the parallel line to BA through D at point E. The segment CE intersects Γ again at F. Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

Problem 3. (Sharygin 2022 Finals 8th Grade P5) An incircle of triangle ABC touches AB, BC, AC at points C_1, A_1, B_1 respectively. Let A' be the reflection of A_1 about B_1C_1, point C' is defined similarly. Lines $A'C_1$ and $C'A_1$ meet at point D. Prove that $BD \parallel AC$.

Problem 4. (Cono Sur 2025 P1) Given a square $ABCD$, let P be a point on the segment BC and let G be the intersection point of AP with the diagonal DB. The line perpendicular to the segment AP through G intersects the side CD at point E. Let K be a point on the segment GE such that $AK = PE$. Let Q be the intersection point of the diagonal AC and the segment KP. Prove that the points E, K, Q, and C are concyclic.

Problem 5. (Tuymaada 2022 Junior P3) Bisectors of a right triangle $\triangle ABC$ with right angle B meet at point I. The perpendicular to IC drawn from B meets the line IA at D; the perpendicular to IA drawn from B meets the line IC at E. Prove that the circumcenter of the triangle $\triangle IDE$ lies on the line AC.

Problem 6. (IGO 2017 Advanced P1) In triangle ABC, the incircle, with center I, touches the sides BC at point D. Line DI meets AC at X. The tangent line from X to the incircle (different from AC) intersects AB at Y. If YI and BC intersect at point Z, prove that $AB = BZ$.

Problem 7. (Baltic Way 2024 P11) Let $ABCD$ be a cyclic quadrilateral with circumcentre O and with AC perpendicular to BD. Points X and Y lie on the circumcircle of the triangle BOD such that $\angle AXO = \angle CYO = 90^\circ$. Let M be the midpoint of AC. Prove that BD is tangent to the circumcircle of the triangle MXY.

Problem 8. (Francophone 2024 Junior P3) Let ABC be an acute triangle with $AB < AC$ and let O be its circumcenter. Let D be a point on the segment AC such that $AB = AD$. Let E be the intersection of the line AB with the perpendicular line to AO through D. Let F be the intersection of the perpendicular line to OC through C with the line parallel to AC and passing through E. Finally, let the lines CE and DF intersect in G. Show that AG and BF are parallel.